

Computational Physics 2 Classes No.8 - Report

2D membrane vibrations (FD version)

Natalia Kowalczyk

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1 Introduction

The aim of this project is to investigate vibrations of a two-dimensional square membrane described by the damped wave equation. The equation is solved numerically using the finite-difference (FD) method on a uniform spatial grid. A periodic external force is applied at the center of the membrane, and the resulting oscillatory motion is studied for different driving frequencies.

Particular attention is paid to the average energy of the membrane in the stationary regime and to the occurrence of resonance phenomena. The influence of the damping coefficient on the position and shape of the resonant peaks is also examined.

2 Results

The finite-difference method was used to solve the damped wave equation on a square membrane with fixed boundary conditions. The membrane was driven by a periodic force applied at the central grid node. Simulations were performed up to $t = 80$, and the average energy was evaluated after the transient stage, i.e. over the interval $10 \leq t \leq 80$.

2.1 Response of the membrane for $\omega = \pi$

To determine whether the membrane reaches a stationary oscillatory state, the system was first simulated for the driving frequency $\omega = \pi$.

Figure 1 shows the displacement of the central node as a function of time. During the first few units of time the amplitude changes noticeably due to transient effects associated with the initial condition $u = 0$. After approximately $t \approx 8$, the oscillations stabilize and become nearly sinusoidal with a constant amplitude. This indicates that the membrane reaches a periodic stationary regime.

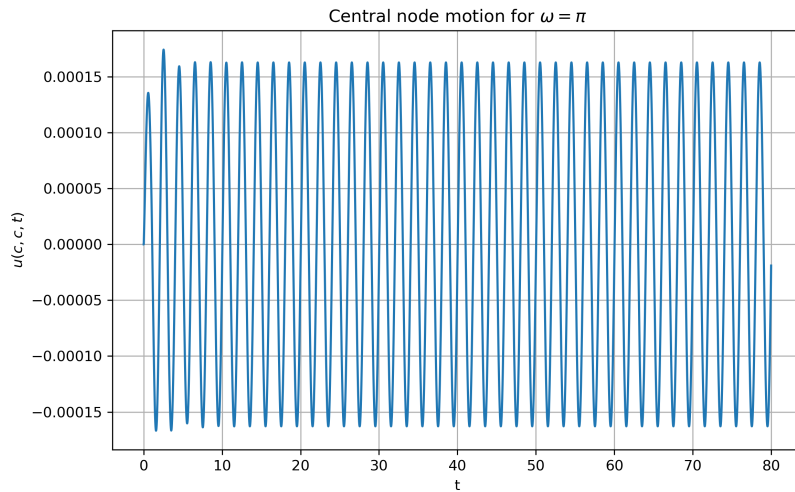


Figure 1: Displacement of the central node as a function of time for $\omega = \pi$.

The evolution of the total energy is shown in Figure 2. Initially the energy increases as the external force transfers energy into the membrane. After the transient stage, the energy oscillates around a constant average value without any systematic growth. This behavior confirms the existence of a stable periodic solution in the presence of damping.

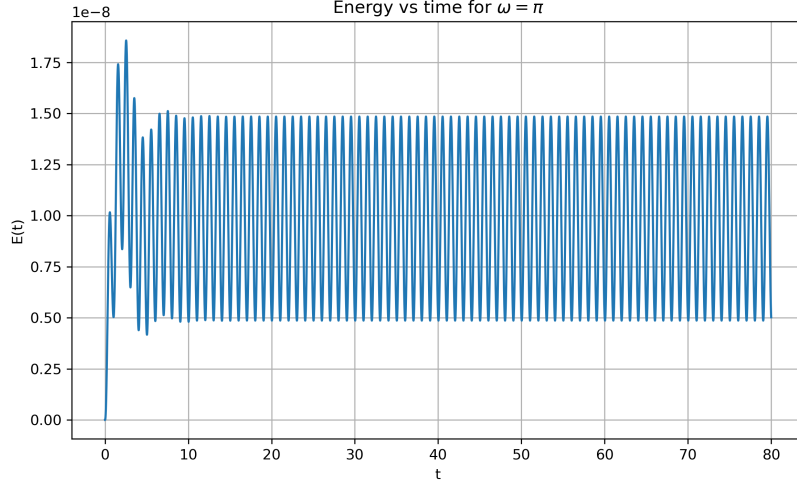


Figure 2: Total membrane energy as a function of time for $\omega = \pi$.

2.2 Average energy as a function of driving frequency

To identify resonant frequencies, the average energy

$$\langle E \rangle = \frac{1}{70} \int_{10}^{80} E(t), dt$$

was calculated for driving frequencies in the range $1.5 \leq \omega \leq 15$.

The resulting dependence is presented in Figure 3. Several pronounced peaks are clearly visible. These peaks correspond to resonant excitation of the membrane eigenmodes, where the driving frequency becomes close to one of the natural frequencies of the system.

The strongest resonance is observed near

$$\omega \approx 10.0.$$

Additional resonances appear near

$$\omega \approx 4.4, \quad \omega \approx 13.3.$$

Between these frequencies the average energy decreases significantly, indicating a less efficient transfer of energy from the external force to the membrane.

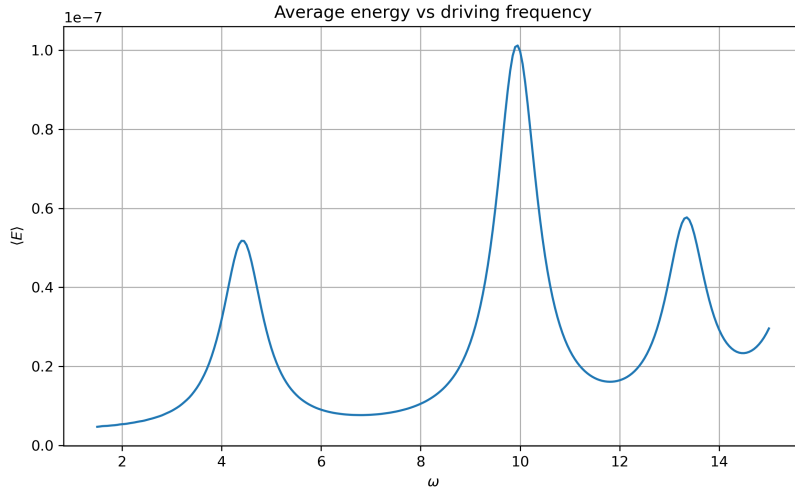


Figure 3: Average membrane energy as a function of driving frequency.

The observed resonances agree well with the theoretical eigenfrequencies of a square membrane,

$$\omega_{mn} = \pi\sqrt{m^2 + n^2},$$

for those modes that can be excited by a force applied at the center of the membrane.

2.3 Influence of damping

The effect of damping was investigated by repeating the frequency scan for several values of the damping coefficient $2d$.

Figure 4 compares the resulting resonance curves. For weak damping, for example $2d = 0.2$, the resonance peaks are very sharp and attain the largest values of the average energy. As the damping coefficient increases, the resonance peaks become lower and broader. This indicates that a larger fraction of the supplied energy is dissipated before it can accumulate in the resonant mode.

An important observation is that the positions of the resonance peaks remain nearly unchanged. Damping mainly affects the peak height and width, while the resonant frequencies themselves are determined primarily by the geometry and boundary conditions of the membrane.

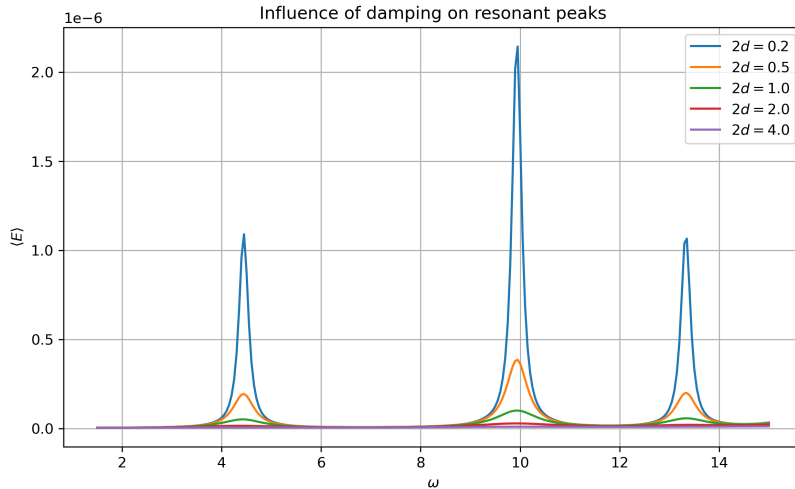


Figure 4: Influence of the damping coefficient on the resonance peaks.

The results are consistent with the general theory of damped driven oscillations. Weak damping leads to strong resonant amplification, whereas strong damping suppresses the response and broadens the resonance curves.

3 Conclusions

In this project, the damped wave equation for a two-dimensional square membrane was solved using the finite-difference method. The numerical simulations showed that for a driving frequency of $\omega = \pi$, the membrane reaches a stationary periodic regime after an initial transient stage. Both the displacement of the central node and the total energy exhibit periodic behavior at long times.

The dependence of the average energy on the driving frequency revealed several resonance peaks, corresponding to the natural vibration modes of the membrane. The strongest resonances were observed near $\omega \approx 4.4$, $\omega \approx 10.0$, and $\omega \approx 13.3$, in good agreement with the theoretical eigenfrequencies of a square membrane.

The influence of damping was also investigated. Increasing the damping coefficient significantly reduced the height of the resonance peaks and broadened their shape, while the resonant frequencies themselves remained nearly unchanged. These results are consistent with the expected behavior of damped driven oscillatory systems.

Overall, the finite-difference approach successfully reproduced the main physical features of membrane vibrations, including stationary oscillations, resonance phenomena, and damping effects.